

Scientific Computation

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Lecture 26 - Numerical Integration

Lecture Outline

- Error in polynomial interpolation
- Chebyshev nodes
- Horner's method

- Introduction to numerical integration
- Trapezoidal rule

Optimally Selected “Nodes”

Suppose $f(x_i)$ is given at $n + 1$ distinct points (or **nodes**) $x_0, x_1, \dots, x_n$ in $[a, b]$ and $p_n(x)$ of degree $\leq n$ interpolates f at these points. We have the following absolute error bound:

$$\max_{x \in [a, b]} |f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \max_{s \in [a, b]} \prod_{i=0}^n |s - x_i|,$$

where $M = \max_{t \in [a, b]} |f^{(n+1)}(t)|$.

How do we choose x_i to minimize this error bound?

Relevant to applications where we can select where to sample $f(x)$.

Note that the x_i 's don't have to evenly spaced.

6

Chebyshev Nodes

Let's assume for simplicity that $[a, b] = [-1, 1]$. Then

$$\min_{x_0, \dots, x_n} \max_{s \in [-1, 1]} \prod_{i=0}^n |s - x_i|$$

is minimized by

decision
variable

$$x_i = \cos \left(\frac{2i+1}{2n+2} \pi \right), i = 0, 1, \dots, n,$$

which are called the **Chebyshev nodes** over $[-1, 1]$.

The minimized value is 2^{-n} (not to be derived in this course)
The Chebyshev nodes are the zeros of Chebyshev polynomials,
which have a rich theory

7

Error Bound with Chebyshev Nodes

The error bound with Chebyshev nodes is therefore

$$\max_{x \in [-1, 1]} |f(x) - p_n(x)| \leq \frac{M}{2^n(n+1)!},$$

where $M = \max_{t \in [-1, 1]} |f^{(n+1)}|$ as usual.

For general $[a, b] \neq [-1, 1]$, we can apply an affine transformation:

$$x_i = \frac{1}{2}(a+b) + \frac{1}{2}(b-a) \cos \left(\frac{2i+1}{2n+2} \pi \right), i = 0, 1, \dots, n.$$

For assignment 3: what is the error bound for Chebyshev nodes on $[a, b]$?

Hint: apply a change of variables to the bound on $x \in [-1, 1]$.

8

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Complexity of Polynomial Evaluation

How many operations are required to evaluate

$$p_n(x) = c_0 + c_1x + c_2x^2 + \dots + c_nx^n? \quad n \text{ additions}$$

$$\begin{array}{l} x^k \Rightarrow k-1 \text{ multiplications} \\ c_i \cdot x^i \Rightarrow +1 \text{ multiplications} \end{array} \quad \left. \vphantom{\begin{array}{l} x^k \Rightarrow k-1 \text{ multiplications} \\ c_i \cdot x^i \Rightarrow +1 \text{ multiplications} \end{array}} \right\} k$$
$$\Rightarrow \sum_{k=1}^n k = \frac{n(n+1)}{2} \Rightarrow O(n^2) \text{ algorithm}$$

Horner's Method

Let's write the polynomial in the following **nested** form:

$$p_n(x) = c_0 + x(c_1 + x(c_2 + \dots + x(c_{n-1} + \underbrace{xc_n}_{b_0} \dots))).$$

What is the complexity now?

$$b_1 = x(c_{n-1} + x c_n)$$

$$b_0 = c_n$$

$$b_1 = c_{n-1} + x c_n = c_{n-1} + x b_0$$

⋮

$$b_n = c_0 + x b_{n-1} = p(x)$$

$\Rightarrow n$ multiplication with n additions

$$\Rightarrow 2n \text{ Flops} \Rightarrow O(n)$$

Lecture Outline

- Chebyshev nodes
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- Introduction to numerical integration

• Theoretical analysis

- trapezoidal rule

The Problem

We seek to **approximately** compute the integral

$$I_f = \int_a^b f(x)dx$$

in situations where a **closed form** solution

- is unknown (maybe it doesn't exist);
- difficult to compute (e.g., the result of a large simulation); or
- is based on data we have collected (e.g., integrating inertial data to track a trajectory).

13

Example from Lecture 1

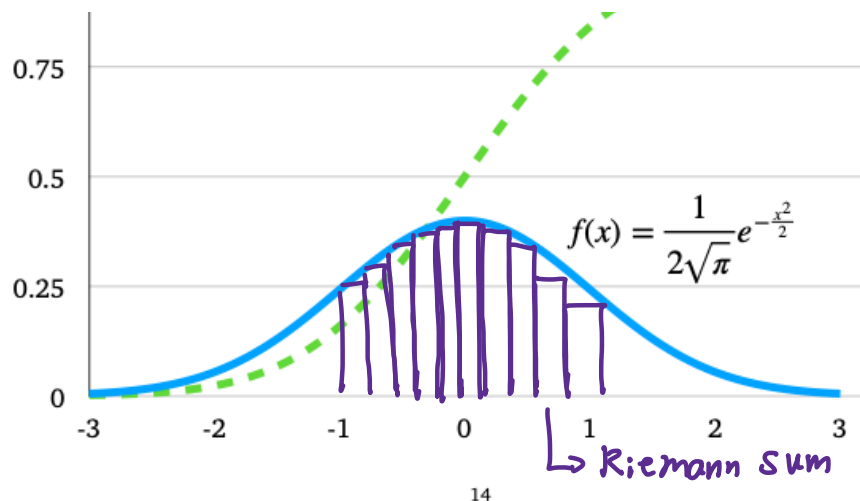
The Cumulative Density Function (CDF) is $F(x) = \int_{-\infty}^x f(t)dt = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$
and has no **elementary** solution (Liouville, 1838)!

— PDF

- - CDF

1



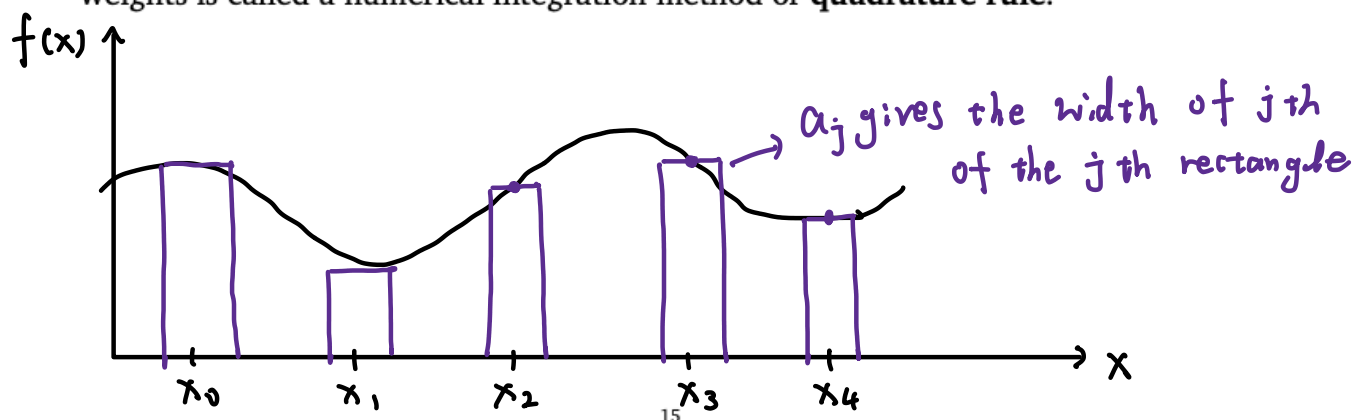


Quadrature Rules

Assume we only know the value of $f(x)$ at a set of $n + 1$ points labelled x_j , for $j = 0, 1, \dots, n$. We use the approximation

$$I_f = \int_a^b f(x) dx \approx \sum_{j=0}^n a_j f(x_j),$$

where the weights a_j are what we must compute. A procedure for determining these weights is called a numerical integration method or **quadrature rule**.



Deriving a Quadrature Rule

Let $x_0, x_1, \dots, x_n$ be distinct points in $[a, b]$ as usual. Let $p_n(x)$ be the interpolating polynomial for $f(x)$ through these points. We know from last week that

$$I_f = \int_a^b f(x) dx \approx \int_a^b p_n(x) dx.$$

From the Lagrange form $p_n(x) = \sum_{j=0}^n f(x_j) L_j(x)$, we have $L_j(x_j) = 1$

Lagrange basis polynomial

$$\begin{aligned} \int_a^b f(x) dx &\approx \int_a^b p_n(x) dx = \int_a^b \sum_{j=0}^n f(x_j) L_j(x) dx \\ &= \sum_{j=0}^n f(x_j) \underbrace{\int_a^b L_j(x) dx}_{a_j} \end{aligned}$$

16

$$L_j(x) = \frac{(x-x_0) \cdots (x-x_{j-1}) \cdot (x-x_{j+1}) \cdots (x-x_n)}{(x_j-x_0) \cdots (x_j-x_{j-1})(x_j-x_{j+1}) \cdots (x_j-x_n)}$$

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The Simplest Case (n=1)

Let $n = 1$ (i.e., we only have the function values on the endpoints $x_0 = a$ and $x = b$). Our Lagrange basis polynomials are therefore

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - b}{a - b}, \quad L_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - a}{b - a}.$$

Using these basis functions,

$$f(x) \approx p_1(x) = f(x_0)L_0(x) + f(x_1)L_1(x)$$

$$\begin{aligned} \text{Integrate } I_f &= \int_a^b f(x) dx \approx f(a) \int_a^b L_0(x) dx + f(b) \int_a^b L_1(x) dx \\ &= f(a) \underbrace{\int_a^b \frac{x-b}{a-b} dx}_{\alpha_0} + f(b) \underbrace{\int_a^b \frac{x-a}{b-a} dx}_{\alpha_1} \\ &= f(a) \int_a^b \frac{x-b}{a-b} dx + f(b) \int_a^b \frac{x-a}{b-a} dx \end{aligned}$$

18

The Simplest Case (n=1) Cont.

Evaluating the integrals gives:

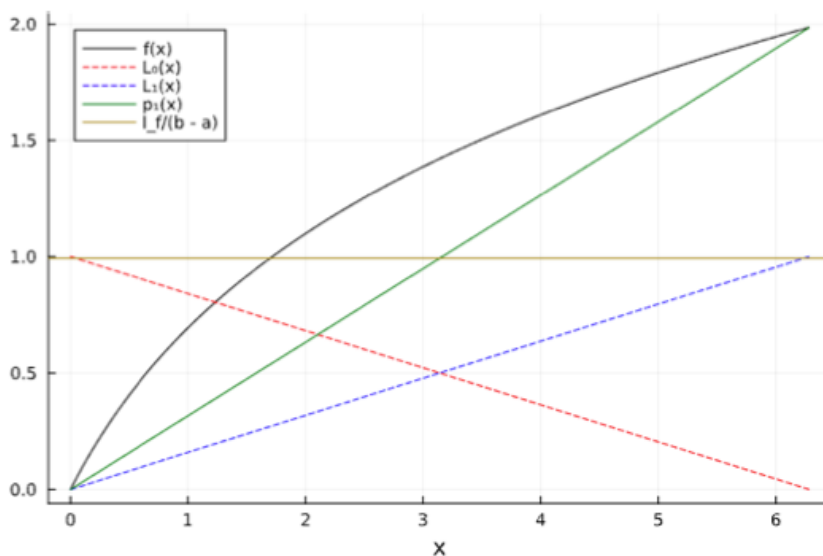
$$\int_a^b \frac{x-b}{a-b} dx = \frac{1}{2} \frac{b^2 - a^2}{a-b} + b$$

$$\int_a^b \frac{x-a}{b-a} dx = \frac{1}{2} \frac{b^2 - a^2}{b-a} - a$$

$$\begin{aligned} \Rightarrow I_f &\approx f(a) \left[b - \frac{b+a}{2} \right] + f(b) \left[-a + \frac{b+a}{2} \right] \\ &= f(a) \left[\frac{2b - (b+a)}{2} \right] + f(b) \left[\frac{b+a - 2a}{2} \right] = \frac{b-a}{2} [f(a) + f(b)] \end{aligned}$$

Visualizing the Result

Plotting the result should make what we've done a little clearer:



```
using Plots
a = 0.0
b = 2*pi

f(x) = log(x+1.0)

x = a:0.01:b

L0(x) = (x - b)/(a - b)
L1(x) = (x - a)/(b - a)

p1(x) = f(a)*L0(x) + f(b)*L1(x)

plot(x, f.(x), label="f(x)", color=:black)
plot!(x, L0.(x), label="L0(x)", color=:red,
       linestyle=:dash)
plot!(x, L1.(x), label="L1(x)", color=:blue,
       linestyle=:dash)
plot!(x, p1.(x), label="p1(x)", color=:green)
hline!([(f(a) + f(b))/2], label="I_f/(b - a)")
xaxis!("x")
```

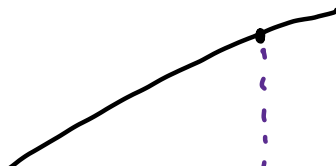
The Trapezoidal Rule

This is called the **trapezoidal rule**:

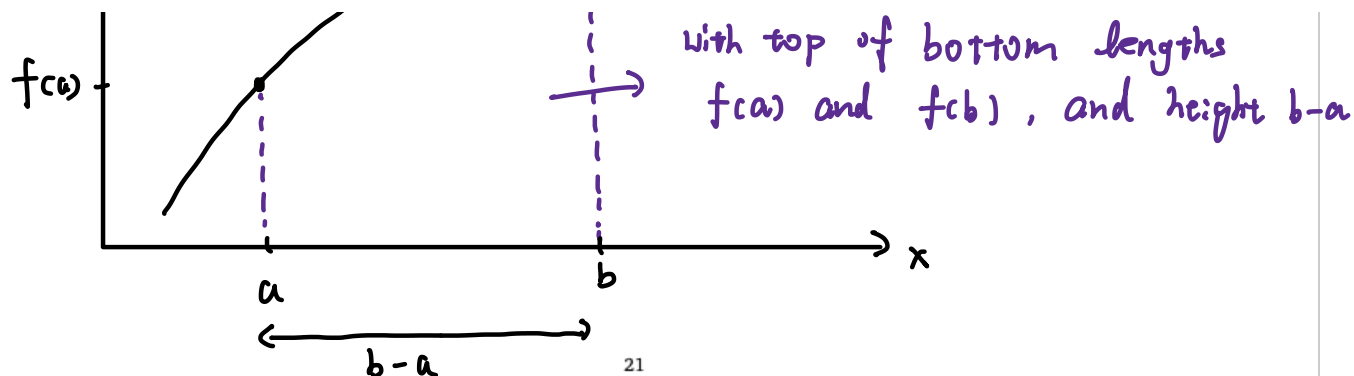
$$I_f \approx I_{\text{trap}} = \frac{b-a}{2} [f(a) + f(b)]$$

We can draw a picture to see why:

$f(x)$
 $f(b)$



Formula for area of a trapezoid



21

Using the Trapezoidal Rule

Example 1: Approximate

$$\int_0^1 e^x dx = e - 1 = 1.7182\dots$$

using the trapezoidal rule.

$$\begin{aligned} I_{\text{trapez}} &= \frac{1}{2} [f(0) + f(1)] \frac{1}{b-a} \\ &= 0.5(e^0 + e^1) \\ &= 1.8591 \end{aligned}$$

22

Using the Trapezoidal Rule Cont.

Example 2: Approximate

$$\int_0^{0.1} e^x dx = e^{0.1} - 1 = 0.010517\dots$$

using the trapezoidal rule.

$$\begin{aligned} I_{\text{trap}} &= \frac{0.1}{2} [f(0) + f(0.1)] \\ &= 0.05 (1 + e^{0.1}) \\ &\approx 0.10525 \end{aligned}$$

23

Next Class: Integration Error

The trapezoidal rule amounts to approximating $f(x)$ by linear interpolation with

$$p_1(x) = f(a) \frac{x-b}{a-b} + f(b) \frac{x-a}{b-a},$$

and then exactly computing the integral of the interpolant. We learned last week that the error of the degree n interpolant is

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i),$$

which for $n = 1$ gives us

24

上一章我们知道了，对于Newton interpolation，插值误差对于均匀取点会爆炸式增长，

Chebyshev定理给出了，按照Chebyshev node来取点，误差最小；

Chebyshev 定理 (Minimax Optimal Nodes Theorem)

在区间 $[-1, 1]$ 上，在所有插值节点选择 $x_0, \dots, x_n$ 的方案中：

$$\min_{x_0, \dots, x_n} \max_{s \in [-1, 1]} \prod_{i=0}^n |s - x_i|$$

的最优值是：

$$2^{-n}$$

并且唯一能够使这个最优值达到的节点就是：

$$x_i = \cos\left(\frac{2i+1}{2n+2}\pi\right), \quad i = 0, 1, \dots, n.$$

我们把这组点叫 **Chebyshev nodes** (切比雪夫节点)。

🔍 定理的含义 (非常重要)

在所有可能的节点中，Chebyshev 节点使误差项 $\prod |x - x_i|$ 的最大值最小。
这就是全局最优、无可替代的节点选择。

也就是说：

- 均匀节点是最差的，会爆炸
- Chebyshev 节点是最优的，会指数收敛

■ 二、为什么这些点是 cos？为什么叫 Chebyshev？

因为它们都是 Chebyshev 多项式 $T_{n+1}(x)$ 的零点。

Chebyshev 多项式满足：

$$T_k(x) = \cos(k \arccos x)$$

因此它的零点在：

$$x_i = \cos\left(\frac{2i+1}{2k}\pi\right)$$

这正是你 slide 上出现的节点公式。

更深层原因：

Chebyshev 多项式有一个惊人的性质：

在所有最高次项系数相同的多项式中，Chebyshev 多项式在 $[-1,1]$ 上的最大振幅最小。

这叫 **最小最大误差性质 (minimax property)**。

Chebyshev 节点继承了这种“最小振荡”性质 → 所以成为最优插值节点。

三、Chebyshev Nodes 的定义

在 $[-1, 1]$ 上：

$$x_i = \cos\left(\frac{2i+1}{2n+2}\pi\right), \quad i = 0, \dots, n$$

在一般区间 $[a, b]$ 上，需要线性变换：

$$x_i = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{2i+1}{2n+2}\pi\right)$$

这就是 实际可用的 Chebyshev 节点公式。

四、使用 Chebyshev 节点时，插值误差是多少？（最重要）

从上一页的误差公式：

$$|f(x) - p_n(x)| = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n |x - x_i|.$$

Chebyshev 定理告诉你：

$$\prod |x - x_i| \leq 2^{-n}.$$

因此误差上界变成：

在 $[-1, 1]$ 上：

$$\max_{x \in [-1, 1]} |f(x) - p_n(x)| \leq \frac{M}{2^n (n+1)!}$$

其中：

$$M = \max_{t \in [-1, 1]} |f^{(n+1)}(t)|.$$

这是课上 slide 的主结论。

在一般区间 $[a, b]$ ：

因为变换时 $(b-a)/2$ 会缩放每个 $|x - x_i|$ ，乘积缩放为：

$$\left(\frac{b-a}{2}\right)^{n+1}$$

所以误差变成：

$$\max_{x \in [a, b]} |f(x) - p_n(x)| \leq \frac{M}{2^n (n+1)!} \left(\frac{b-a}{2}\right)^{n+1}$$

这就是你作业 assignment hint 要你推导的结果。

■ 五、最终一句话总结

Chebyshev 定理说：Chebyshev nodes 是最优插值节点，它们能够最小化最坏情况下的误差，而且误差收敛速度为 $O(2^{-n}/(n+1)!)$ ，远比均匀节点稳定得多。

第二部分讲了 Horner method，原本计算一个多项式要 $O(n^2)$ ，用了这个方法直接变成 $O(n)$ 。